Notes:

i. All questions are compulsory.
ii. Figures to the right indicate full marks.
iii. Graph paper is necessary for L.P.P.
iv. Use of logarithmic table is allowed.
v. Answers to the question in Section – I and Section – II should be written in two separate answer books.
vi. Question from Section – I attempted in the answer book of Section – II and vice-versa will not be assessed / not be given any credit.
vii. Answer to every question must be written on a new page.

Section – I

Q.1. Attempt any SIX of the following: **[12]**

i. p: It is a day time, q : It is warm
Give the verbal statements for the following symbolic statements :
a. $p \wedge \sim q$ b. $p \rightarrow q$ (2)

ii. Express the truth of each of the following statements using Venn diagrams:
a. No circles are polygons
b. Some quadratic equations have equal roots. (2)

iii. Find the values of x and y if
$$2\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix}+\begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix}=\begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$ (2)

iv. Find $\frac{dy}{dx}$; if $x = \sin^3 \theta$, $y = \cos^3 \theta$ (2)

v. Find $\frac{dy}{dx}$; if $y = \cos^{-1}\left(2x\sqrt{1-x^2}\right)$ (2)

vi. Evaluate: $\int x \cdot \log x \, dx$ (2)

vii. The cost C of producing x articles is given as $C = x^3 - 16x^2 + 47x$. For what values of x, the average cost is decreasing? (2)

viii. Evaluate: $\int_0^{\frac{\pi}{4}} \frac{1}{1+x^2} dx$ (2)

Q.2. (A) Attempt any TWO of the following: **(6)[14]**

i. Solve the following equations by reduction method:
$x + y + z = 6$, $3x - y + 3z = 10$, $5x + y - 4z = 3$ (3)

ii. Evaluate : $\int \frac{2x+1}{(x+1)(x-2)} dx$ (3)

iii. Evaluate : $\int_0^1 x\,(1-x)^{\frac{3}{2}} dx$ (3)

(B) Attempt any TWO of the following: **(8)**

i. Using the rules of negation, write the negation of the following :

a. $p \wedge (q \rightarrow r)$ (b) $\sim p \vee \sim q$ (4)

ii. If the function f is continuous at $x = 2$ and $x = 4$ then find the values of a and b.

Where $f(x) = x^2 + ax + b, \; x < 2$

$= 3x + 2, \quad 2 \leq x \leq 4$

$= 2ax + 5b, \quad 4 \leq x$ (4)

iii. A manufacturing company produces x items at the total cost of ₹ $(180 + 4x)$. The demand function of this product is $P = (240 - x)$ Find x for which profit is increasing. (4)

Q.3. (A) Attempt any TWO of the following: **(6)[14]**

i. If $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 7 & 1 \\ 2 & 5 \end{bmatrix}$, verify that $|AB| = |A|.|B|$ (3)

ii. Evaluate : $\int \frac{1}{x.[(\log x)^2 + 4]} dx$ (3)

iii. Find the volume of solid generated by rotating the area bounded by $x^2 + y^2 = 36$ and the lines $x = 0, x = 3$ about X – axis. (3)

(B) Attempt any TWO of the following: **(8)**

i. If f is continuous at $x = 0$, then find $f(0)$. Where $f(x) = \frac{(3^{\sin x} - 1)^2}{x.\log(x + 1)}, x \neq 0$ (4)

ii. If $x^7 . y^9 = (x + y)^{16}$, then show that $\frac{dy}{dx} = \frac{y}{x}$ (4)

iii. In a firm the cost function for output x is given as $C = \frac{x^3}{3} - 20x^2 + 70x$. Find the output for which marginal cost (C_m) is minimum. (4)